

BUS 304: ANALYSIS FOR BUSINESS DECISIONS TRANSPORTATION MODEL — PRACTICE QUESTION BANK

60 Questions | 20 Theory | 20 Fill-in-the-Gap | 20 MCQ | OAU Standard

SECTION A: THEORY QUESTIONS (20 Questions)

Instructions: Answer all questions. Provide detailed, well-structured responses. Marks are awarded for accuracy, depth of explanation, and correct use of terminology.

1. Distinguish between a **balanced** and an **unbalanced** transportation problem. Explain the step-by-step procedure for converting an unbalanced problem into a balanced one, and state precisely why this conversion is compulsory before applying any solution method.
2. Critically compare the **North West Corner Rule (NWCR)**, **Least Cost Method (LCM)**, and **Vogel's Approximation Method (VAM)** with respect to: (a) the basis of allocation, (b) quality of initial solution obtained, and (c) computational effort required. Which method would you recommend to a project manager and why?
3. Define **degeneracy** in the context of the transportation model. State the mathematical condition for a non-degenerate basic feasible solution and explain in detail why a degenerate solution cannot be tested for optimality using the Stepping Stone Method.
4. A transportation problem has **4 origins** and **5 destinations**. (a) Calculate the minimum number of occupied cells required for a non-degenerate basic feasible solution, showing your workings. (b) If the solution contains only 7 occupied cells, identify the problem and state exactly how it is resolved.
5. Explain in detail the procedure for computing **improvement indices** using the Stepping Stone Method. In your answer, state all the rules governing the tracing of a closed path. What does a **negative improvement index** signify in a minimisation problem, and what action does it require?
6. State ALL four **basic requirements** and ALL four **assumptions** of the transportation model. For each of the four requirements, explain the specific consequence to the model if that requirement is not satisfied.
7. In Vogel's Approximation Method, what is an **opportunity cost (penalty)** and how is it computed for a row and a column? Explain in full how ties are resolved when: (a) two or more rows/columns have equal opportunity costs, and (b) two or more cells in the selected row/column have equal unit costs. Provide the rationale behind each tie-breaking rule.
8. Using the concept of improvement indices, explain precisely how you determine that an optimum solution has been reached in: (a) a **cost-minimisation** transportation problem, and (b) a **profit-maximisation** transportation problem. Why do the conditions differ between the two cases?

9. Explain the **Rim Condition** in the transportation model. If the total supply of a company is **650 units** and total demand is **420 units**, describe the exact steps you would take to balance this problem. In a business context, what does an allocation to a dummy destination represent?
10. State ALL the rules governing the tracing of a **closed path** in the Stepping Stone Method. Explain what happens when it is impossible to trace a closed path for an empty cell, and what this condition indicates about the current solution.
11. "VAM gives the closest approximation to the optimum solution among the three basic solution methods." Justify this statement by explaining the **theoretical basis** of VAM's superiority. Why does the use of penalty costs lead to better initial allocations compared to NWC and LCM?
12. A company has **three factories** and **four markets**. After applying the Least Cost Method, the number of allocated cells is **5**. (a) Using the appropriate formula, determine whether this solution is degenerate or non-degenerate. Show all workings. (b) What is the direct implication for applying the Stepping Stone Method?
13. Explain the role of a **dummy variable** in an unbalanced transportation problem. (a) What unit cost is assigned to all cells in the dummy row or column, and why? (b) When supply exceeds demand, is a dummy row or dummy column introduced? (c) What does a non-zero allocation to a dummy destination represent in real business terms?
14. Discuss the transportation model as both a **cost-minimisation** and a **profit-maximisation** tool. Under what specific business circumstances would a firm use it for profit maximisation rather than cost minimisation? What changes in the optimality condition and interpretation of improvement indices would result?
15. Distinguish between a **basic feasible solution** and an **optimum solution** in a transportation model. Is it theoretically possible for the initial basic feasible solution obtained from VAM to be the optimum solution? Explain your answer fully.
16. In the Stepping Stone Method, after identifying the closed path for the cell with the most negative improvement index: (a) Explain how the alternating **+/-** signs are assigned along the closed path. (b) What is the significance of selecting the **smallest negative-signed allocation**? (c) What happens to that cell (which held the smallest negative-signed allocation) after the reallocation is completed?
17. A transportation solution with **R rows** and **C columns** satisfies the formula $R + C - 1$ occupied cells, but one of the occupied cells has a **zero allocation**. (a) Is the solution degenerate or non-degenerate? Justify your answer. (b) State the implication of this condition for the Stepping Stone Method.
18. Explain the **Least Cost Method (LCM)** step by step. In your answer: (a) state the specific rule applied when there is a tie in unit transportation costs; (b) explain how the smaller allocation unit is determined at each step; (c) identify the fundamental difference between LCM and the NWC Rule in their approach to allocation.
19. State the **advantages and limitations** of the Stepping Stone Method. Under what conditions does SSM become particularly tedious and impractical? What does it mean when an improvement

index equals **zero** for an empty cell in an otherwise optimal solution?

20. "The transportation model can only be applied when the Rim Condition is fully satisfied." Discuss this statement comprehensively. In your answer, explain: (a) what the Rim Condition entails and why it is essential to the mathematical structure of the model; (b) how the condition is enforced when the natural problem violates it; (c) the consequence of attempting to solve an unbalanced problem without first balancing it.

SECTION B: FILL IN THE GAP (20 Questions)

Instructions: Fill in each blank with the most appropriate word(s) or phrase. Each blank carries equal marks. Spelling and technical accuracy are required.

1. The transportation model is a mathematical technique that involves developing an effective plan to distribute goods from several points of _____ to a number of points of _____.

Answer: supply (origins) ; demand (destinations)

2. The condition which requires that total quantity supplied must be equal to total quantity demanded in a transportation model is known as the _____ Condition.

Answer: Rim

3. In the transportation model, items to be transported are assumed to be _____, meaning they are the same regardless of their source or destination.

Answer: homogeneous

4. The three methods used to obtain an initial basic feasible solution in a transportation problem are _____, _____, and _____.

Answer: North West Corner Rule ; Least Cost Method ; Vogel's Approximation Method

5. In the North West Corner Rule, allocation begins at the _____ cell of the transportation tableau.

Answer: upper left-hand (top-left / north-west)

6. The Least Cost Method allocates units based on _____ transportation cost, and is described as efficient and _____.

Answer: least (lowest) ; cost-effective

7. In VAM, the opportunity cost (penalty) for each row and column is calculated as the difference between the _____ and the _____ lowest transportation cost in that row or column.

Answer: second lowest ; lowest

8. In Vogel's Approximation Method, if exactly one row or column remains uncrossed, VAM is stopped and _____ is used to complete the remaining allocation.

Answer: Least Cost Method (LCM)

9. In VAM, when there is a tie in opportunity cost (penalty) figures, allocation is made to the cell that has the _____ least unit cost.

Answer: smaller

10. In VAM, when there is a tie in unit cost figures, allocation is made to the cell that can take _____ allocation.

Answer: greater

11. An unbalanced transportation problem arises when the total supply at the origins does _____ the total demand at the destinations.

Answer: not equal

12. To convert an unbalanced transportation problem into a balanced one, a _____ source or destination is introduced with _____ transportation cost.

Answer: dummy ; zero

13. The optimum solution of a transportation model is defined as the approach that results in the _____ possible total transportation cost.

Answer: lowest

14. For a **minimisation** problem, the optimum solution is obtained when all improvement indices are either _____ or _____.

Answer: zero ; positive

15. For a **maximisation** problem, the optimum solution is obtained when all improvement indices are either _____ or _____.

Answer: zero ; negative

16. The Stepping Stone Method is used to evaluate the economics of transporting via routes that are _____ currently part of the basic solution.

Answer: not

17. The formula for computing the required number of stone (occupied) cells for a non-degenerate transportation solution is: Number of stone cells = _____.

Answer: $R + C - 1$ (Number of rows + Number of columns - 1)

18. In the Stepping Stone Method, when tracing a closed path, movement is permitted only _____ and _____ — diagonal movement is strictly not allowed.

Answer: horizontally ; vertically

19. In the Stepping Stone Method applied to a minimisation problem, the empty cell selected for entry into the solution must have a _____ improvement index.

Answer: negative

20. In Step 6 of the Stepping Stone Method, the _____ allocation among all negative-signed cells on the closed path is deducted from negative cells and added to positive cells.

Answer: smallest

SECTION C: MULTIPLE CHOICE QUESTIONS (20 Questions)

Instructions: Choose the ONE most correct answer for each question. No partial marks for multiple selections.

1. Which of the following is **NOT** a basic requirement of the transportation model?

- (a) A list of origins with their corresponding supply quantities
- (b) The unit profit of each transportation route
- (c) The unit cost of transporting items from each origin to each destination
- (d) Total quantity supplied must equal total quantity demanded

Answer: b — The model requires unit costs, not unit profits, as a basic requirement.

2. The Rim Condition in a transportation model states that:

- (a) Total transportation cost must be minimised
- (b) Total supply must equal total demand
- (c) All routes must be used at least once
- (d) The number of rows must equal the number of columns

Answer: b — The Rim Condition is the equality of total supply and total demand.

3. Which basic solution method **completely ignores** transportation costs during allocation?

- (a) Least Cost Method
- (b) Vogel's Approximation Method
- (c) North West Corner Rule
- (d) Stepping Stone Method

Answer: c — NWC allocates based solely on position (top-left to bottom-right), never on cost.

4. In VAM, the opportunity cost (penalty) for a row is calculated as:

- (a) The lowest cost minus the highest cost in that row
- (b) The average of all unit costs in that row
- (c) The second lowest cost minus the lowest cost in that row
- (d) The highest cost minus the second highest cost in that row

Answer: c — $\text{Penalty} = 2\text{nd lowest cost} - \text{lowest cost in each row/column}$.

5. A transportation problem has **3 origins** and **4 destinations**. The number of stone cells required for a non-degenerate basic feasible solution is:

- (a) 12
- (b) 7
- (c) 6
- (d) 5

Answer: c — $R + C - 1 = 3 + 4 - 1 = 6$ stone cells.

6. Which of the three basic solution methods gives the **closest approximation** to the optimum solution?

- (a) North West Corner Rule

- (b) Least Cost Method
- (c) Vogel's Approximation Method
- (d) All three methods give the same result

Answer: c — VAM uses penalty costs to guide allocation, producing results nearest to the optimum.

7. When applying the Stepping Stone Method to a **minimisation** problem, the empty cell selected for improvement must have:

- (a) The highest positive improvement index
- (b) A zero improvement index
- (c) The most negative improvement index
- (d) Any index value — the choice is arbitrary

Answer: c — The most negative index indicates the greatest potential for cost reduction.

8. A transportation problem has total supply of **800 units** and total demand of **600 units**. To balance the problem, you should:

- (a) Add a dummy origin (row) with supply of 200 and zero costs
- (b) Add a dummy destination (column) with demand of 200 and zero costs
- (c) Reduce the supply of the largest origin by 200
- (d) Increase the demand of the largest destination by 200

Answer: b — Excess supply ($800 > 600$) means demand is short; add a dummy column with demand = 200.

9. In the Least Cost Method, when there is a **tie in unit transportation costs** between two or more cells:

- (a) The row with the higher supply is selected first
- (b) The column with the higher demand is selected first
- (c) Allocation can be made to any of the tied cells
- (d) VAM must be used to break the tie

Answer: c — LCM permits allocation to any tied cell at the solver's discretion.

10. The optimum solution in a **profit-maximisation** transportation problem is reached when:

- (a) All improvement indices are strictly positive
- (b) All improvement indices are zero or negative
- (c) All improvement indices are zero or positive
- (d) At least one improvement index is negative

Answer: b — For maximisation, all indices must be zero or negative for optimality.

11. In VAM, when exactly **one row or column remains uncrossed** after allocations, the correct action is to:

- (a) Continue applying VAM to that remaining row/column
- (b) Use the North West Corner Rule to complete allocations
- (c) Use the Least Cost Method to complete the remaining allocations
- (d) Stop and declare the basic solution complete as-is

Answer: c — VAM's rule: if exactly one row/column remains, switch to LCM to complete allocation.

12. Which of the following is a stated **assumption** of the transportation model?

- (a) Transportation cost per unit varies with quantity transported on a route
- (b) Items to be transported may be heterogeneous
- (c) There may be multiple routes between any origin-destination pair
- (d) Transporting cost per unit on a route is constant regardless of quantity transported

Answer: d — Constant per-unit cost on any route is a core assumption of the model.

13. In the Stepping Stone Method, the closed path for an empty cell must:

- (a) Pass through every occupied cell in the entire tableau
- (b) Move diagonally through occupied cells where necessary
- (c) Move only horizontally and vertically through occupied cells, returning to the start
- (d) Include at least two unoccupied cells in addition to the starting cell

Answer: c — Closed paths use only horizontal and vertical moves through occupied cells.

14. If a transportation solution has **fewer** stone cells than $R + C - 1$, the solution is classified as:

- (a) Non-degenerate
- (b) Optimal
- (c) Degenerate
- (d) Infeasible

Answer: c — Fewer stone cells than $R + C - 1$ defines degeneracy.

15. In VAM, when two rows or columns are tied on opportunity cost, the tie is broken by making allocation to the cell with:

- (a) The larger allocation capacity
- (b) The smaller least unit cost
- (c) The highest unit cost in the tied row/column
- (d) The cell nearest to the top-left corner of the tableau

Answer: b — Tie in penalty: allocate to the cell with the smaller least unit cost.

16. The North West Corner Rule derives its name from:

- (a) The direction in which unit costs are compared during allocation
- (b) The starting position at the top-left (north-west) cell of the tableau
- (c) The geographic position of the supply sources relative to destinations
- (d) The compass direction in which goods physically move

Answer: b — 'North West' refers to the top-left starting cell of the cost matrix.

17. After computing a new transportation cost in the Stepping Stone Method, what determines whether another round of improvement is necessary?

- (a) Whether the new total cost is lower than the previous total cost
- (b) Whether any improvement index for an empty cell is still negative (for minimisation)
- (c) Whether any improvement index for an empty cell is still positive
- (d) Whether the distribution plan has changed from the previous iteration

Answer: b — Another round is required as long as any empty cell has a negative improvement index.

18. The total transportation cost on any route in the transportation model equals:

- (a) Unit cost divided by the number of units transported on that route

- (b) Number of units transported plus the unit cost on that route
- (c) Number of units transported multiplied by the unit cost on that route
- (d) Unit cost multiplied by the total number of routes in the network

Answer: c — Total route cost = units transported x unit cost (as stated in the assumptions).

19. In the Stepping Stone Method Step 6, the **quantity to be reallocated** is determined by:

- (a) The largest allocation among the positive-signed cells on the closed path
- (b) The smallest allocation among the negative-signed cells on the closed path
- (c) The arithmetic average of all allocations on the closed path
- (d) The total supply of the origin row involved in the closed path

Answer: b — The smallest negative-signed allocation limits reallocation to prevent negative values.

20. The transportation model is described as both a **cost-minimisation** and a **profit-maximisation** tool because:

- (a) It simultaneously minimises cost and maximises profit in every application
- (b) Depending on the nature of the problem, it can be configured to minimise cost or maximise profit/sales
- (c) It maximises total cost while minimising per-unit profit across all routes
- (d) All transportation problems are profit-maximisation problems by mathematical design

Answer: b — The model is flexible; objective function depends on the business decision context.